

NUCLEAR & CLEAN-ENERGY INNOVATION

Opportunity Intelligence Platform

Scoring, ranking, and building a business case for 14 emerging technologies relevant to nuclear innovation and business development.

Karan Gill | Portfolio Project | July 2026

Independent research project — not affiliated with, commissioned by, or reviewed by any employer named within.

How the Platform Works

Four stages, one traceable chain from public source to recommendation

1

Research

14 technologies profiled from public sources (IAEA, OECD NEA, CNSC, DOE, NWMO). Every field tagged VERIFIED / ESTIMATED / ASSUMPTION.

2

Score

Transparent weighted model, 10 criteria, adjustable weights, full sensitivity analysis on every weight.

3

Synthesize

TF-IDF + KMeans clustering and semantic search over a real, source-linked research corpus — 100% cluster-to-category validation.

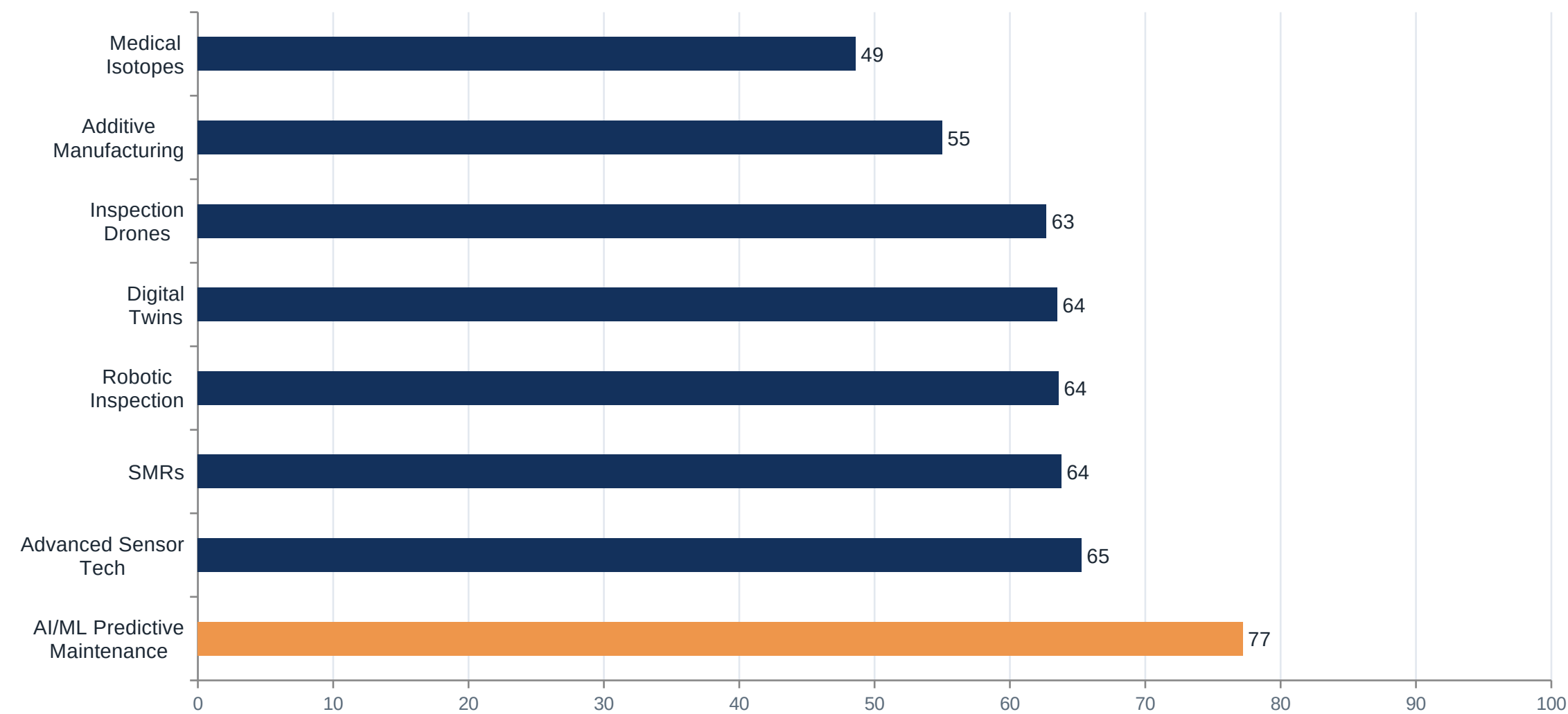
4

Recommend

Ranked dashboard, business case, and stage-gated roadmap for the top opportunity.

Ranked Opportunity Portfolio

Composite score (0-100), weighted model — top 8 of 14 technologies shown



Recommendation

AI/ML-Based Predictive Maintenance — highest-ranked, rank-stable under sensitivity testing

77.2

Composite score
(highest of 14)

Low

Implementation
cost tier

5 / 5

Technical
feasibility rating

2 / 5

Regulatory
complexity (lower = better)

GO — proceed to a scoped proof-of-concept

- Condition: confirm 12-24 months of historical maintenance/sensor data for one equipment class before committing model-development time.
- Phased path: research & validation → proof of concept → shadow-mode pilot → scale, each gated on the prior stage's success criteria.
- Full detail: [business_case_ai_predictive_maintenance.docx](#) and [innovation_roadmap.md](#).

What This Demonstrates

Mapped directly to the Strategic Growth & Innovation Co-op role

1

Investigate & synthesize

14-technology public-source dataset with full confidence labeling (VERIFIED / ESTIMATED / ASSUMPTION).

2

Market & competitive analysis

Market assessment grounded in named, checkable sources — including a real Kinectrics-adjacent fact (the Isogen medical-isotope joint venture).

3

Applied AI/ML

NLP pipeline solving a real triage problem, validated at 100% cluster-to-category accuracy — not added for its own sake.

4

Business case & roadmap

Full CAPEX/OPEX framing, stage-gated roadmap, risk register, and an honest go/no-go call with a stated open condition.